APM348

Mathematical Modelling

University of Toronto **Fall 2026**

This course offers an overview of different approaches for representing physical situations mathematically, followed by analytical techniques and numerical simulations to gain insight into those situations. We will explore questions from biology, economics, engineering, medicine, physics, physiology, and the social sciences, formulating them as problems in optimization, differential equations, and probability.

How this class works

Mathematics is not a spectator sport, and our class will reflect this. Using Inquiry-Based Learning, you will regularly work in pairs to present your team's solutions and strategies to real-world modelling problems during both lectures and tutorials.

This structure reflects how applied science actually happens: through trial and error, presenting to colleagues, discussion, and collaborative refinement. While stepping up to the board might feel intimidating at first, you will solidify your understanding of the course material by articulating it to others, defending your ideas, and collaborating with your peers to deduce new insights.

Embrace the challenge, learn from both successes and missteps, and get ready to take true ownership of your mathematical journey!

Learning Objectives

After this course you will learn the following:

- **Modelling.** To apply many common modelling approaches to a variety of real-world scenarios.
- **Analysis.** To develop and analyze mathematical models.
- **Computer approximations.** To use a computer to approximate solutions and explain advantages and disadvantages of computer-based approximations.
- **Scientific communication.** To write a scientific report where you model a real-world scenario and analyze your model using appropriate techniques.
- **Research.** To search and learn from research literature.

To Succeed

Learning is hard! It is exercise for the mind, and like exercise, when you're doing it, it feels pretty uncomfortable. Here are some tips to help you succeed academically (getting the grade you want) and intellectually (learning the most you can).

- Form a regularly-meeting study group of 3–4 people. Math should not be done in isolation! You need others to bounce ideas off of and to motivate you when you're feeling down.
- Read the class slides and try to solve the exercises before class. A good rule of thumb is every hour spent studying before class is worth two hours of studying after class.
- Force yourself to explain. When reviewing, it's easy to glance at a solution and think,"Oh yeah, I knew that." Don't do it! Force yourself to explain each problem/concept without referencing a solution. Be patient with yourself — it might take 5 or 10 minutes before it finally comes to mind, but studying this way will be significantly more effective.

Prerequisites

- Linear Algebra (MAT223): matrix algebra (rank, nullity, null space), bases, solving linear systems, eigenvalues and eigenvectors.
- Ordinary Differential Equations (MAT244): modelling with odes, separable odes, Euler's method, systems of autonomous odes, stability of equilibrium solutions via eigenvalues, linearization near an equilibrium solution.
- Probability and Statistics (STA237): continuous probability density and distribution functions, mean, conditional probabilities, hazard function, exponential random variable.

If you're feeling shaky on either of those topics, start reviewing earlier rather than later.

Grading Scheme

Start Here	*2%*	Early assignment to review some basic skills on LaTeX and python.
Presentations	*28%*	Each student will give a minimum of 4 presentations in lectures and tutorials throughout the term.
Two Midterms	*30%*	There will be two term tests. The lowest will one will be worth 10% and the highest one 20%.
Final Project: *Outline (5%)* *Meeting (5%)* *Presentation (15%)* *Report (15%)*	*40%*	In teams of 2 or 3, students will produce an outline, meet with the instructor to discuss the outline, give a presentation on the last week of classes, ask questions for each presentation, and will produce a typed final report. (All steps of this Final Project are mandatory.)
Participation	$P \in [0,1]$	Each student will actively participate and ask questions to the presenters during lectures and tutorials. See below for more details.

Final Grade: Your course mark will be computed as a weighted average of the above components and *then it will be multiplied by your participation score P*.

Participation

Participation in class is essential for learning in APM348. It may take many forms, including sharing during class, asking good questions, contributing to group work, or actively engaging and working on the day's activities. Your participation will be summarized by a participation score $P \in [0,1]$.

It will be computed in the following way, let $k \in [0,1]$ be the proportion of lectures and tutorials you participated in. Your participation score will be $P = \min\left(\frac{k}{0.8}, 1\right)$. That is, if you participate in 80% or more of lectures and tutorials, your participation score will be 1.0, and the score scales down linearly from there.

Presentations

Throughout the semester, each student will present at least four times.

- These presentations will be done in groups of 2 students. The groups will change with each presentation – each student will not work with the same partner twice.
- There must be at least one presentation about each of the three main class topics (optimization models, dynamical models, and probability models).
- There must be at least one presentation in tutorial.

Students must be able to explain, justify, and defend all aspects of their work. During presentations, students will be expected to demonstrate a clear understanding of their models, code, results, and conclusions.

Midterm Tests

There will be two midterms, the midterms will be written during 2h lectures and are tentatively scheduled for ***Wednesday, October 6 and November 17***.

Your grade for the tests will be calculated as follows:

- Each student's lowest test will be worth 10% of the course grade;
- Each student's highest test will be worth 20% of the course grade;

The midterms will have two stages: an initial individual part and then a second group part.

The group part will be written in groups of 3 students and will include programming, so each group will need to bring one laptop.

Final Project

Students should work in teams of two or three.

For the project, you can develop a novel mathematical model related to your topic, or provide extensions or new applications for an existing mathematical model. It will not be sufficient to study an existing model and present already reported results for the model; there must be a novel component to your project.

The Final project will have **four mandatory** parts.

1. **Outline.** You will submit an outline for your final project on ***Tuesday, October 20***. Start thinking early about topics that interest you and locate research materials on that topic. No extensions to this deadline will be given.
2. **Meeting.** You will schedule a meeting with the instructor during the week of **Oct 21 – 23 or Nov 2 – Nov 6** to discuss the outline and the project.
3. **Presentation.** You will give a 10 minute presentation in lecture or tutorial during the *last week of classes*. Attendance at the presentations of other groups is mandatory and will form part of your mark.
4. **Report.** You will submit a typed research report which will be due on ***Tuesday, December 8***. No extensions to this deadline will be given.

Missed Assessments

Participation. We recognize that illness and other unforeseen circumstances may occasionally prevent you from attending class. The grading scheme accounts for this by requiring attendance at 80% of lectures and tutorials. This built-in flexibility is intended to accommodate occasional missed classes, and individual adjustments to the participation mark will not normally be provided.

Term Tests. There will be no make-up tests and unexcused missed exams will be given a score of 0. If you miss an exam due to illness, you must submit (via Quercus) a scanned copy of a completed University of Toronto Verification of Student Illness or Injury form (http://www.illnessverification. utoronto.ca) **no later than one week after the test**.

If you find yourself in a situation where you will miss a large portion of the semester and/or assignments, please contact your college registrar to discuss accommodation options.

Classes

There is only one lecture section and one tutorial section.

Time	Room	Instructor	Email	Office
T9-11	MY315	Prof. Bernardo Galvão-Sousa	`beni@math.toronto.edu`	PG114
F12-1	MY420	Prof. Bernardo Galvão-Sousa	`beni@math.toronto.edu`	PG114

Time	Room	Teaching Assistant	Email
F2-3	MY420	Delaram Sadatamin	`delaram.sadatamin@mail.utoronto.ca`

- Tutorials will begin Friday, September 11.
- Attendance in lectures and tutorials is mandatory.
- During lectures and tutorials, students will be presenting solutions to problems to the other classmates.
- When not presenting, you are expected to actively participate and ask questions – your participation will make up part of your grade.
- Please bring a computer (e.g. a laptop or tablet preferably with keyboard) to lectures and tutorials. We will be using Python through `https://utoronto.syzygy.ca/`.

Textbooks

- "Math Modelling: Getting Started and Getting Solutions" by K. M. Bliss, K. R. Fowler, and B. J. Gallizzo, SIAM 2014. `https://m3challenge.siam.org/resources/modeling-handbook`
- "A Course in Mathematical Biology – Quantitative Modeling with Mathematical and Computational Methods" by Gerda de Vries, Thomas Hillen, Mark Lewis, Johannes Muller, and Birgitt Schonfisch. SIAM 2006. ISBN-13: 978-0898716122, `https://www.math.ualberta.ca/~thillen/book-page.html`
- "Computational Mathematical Modeling: An Integrated Approach Across Scales" by Daniela Calvetti and Erkki Somersalo ISBN-13: 978-1611972474, `https://epubs.siam.org/doi/book/10.1137/1.9781611972481`
- "Mathematical Modeling" by Mark M. Meerschaert (4th ed.) ISBN-13: 978-0123869128, `https://www.sciencedirect.com/book/9780123869128/mathematical-modeling`
- "Modeling and Simulation in Medicine and the Life Sciences" by Frank C. Hoppensteadt and Charles Peskin (2nd ed.) ISBN-13: 978-0387215716, `https://link.springer.com/book/10.1007/978-0-387-21571-6`

- Modelling with "Differential Equations" by B. Galvão-Sousa, J. Siefken, `https://siefkenj.github.io/IBLODEs`
- Lecture Notes from APM348 - Winter 2024 by Adam Stinchcombe, `https://www.overleaf.com/read/knncdtfvrzph`

Email & Etiquette

We will try to respond to emails as soon as possible, but during busy times (like before an exam) it might take several days to respond. If your situation is urgent, talk to a professor after class or in office hours.
When writing an email:

- **Identify your email.** Put APM348 in the subject line, use your `utoronto.ca` email, and identify yourself by name and UTORid.
- **Be specific.** We're better able to help you if you're specific about your issue and you include all necessary information. If your situation is complex, it is best to come to office hours to discuss it.
- **Be professional.** Please use appropriate tone and level of formality in your emails. Do not use slang or texting abbreviations. It is tradition in North America to start emails *Dear Professor* . . . , and end them *Thank you,* . . .
- **Check the syllabus and course webpage first.** If your question is answered on the syllabus or the course webpage, we may not respond to your email.
- **No content questions.** If you have mathematical questions, please bring your question to office hours.

Academic Resources

Accessibility Needs. The University provides academic accommodations for students with disabilities in accordance with the terms of the Ontario Human Rights Code. This occurs through a collaborative process that acknowledges a collective obligation to develop an accessible learning environment that both meets the needs of students and preserves the essential academic requirements of the University's courses and programs.
Students with diverse learning styles and needs are welcome in this course. If you have a disability that may require accommodations, please feel free to approach your Course Instructor and/or the Accessibility Services office as soon as possible. The sooner you let us know your needs the quicker we can assist you in achieving your learning goals in this course.

- `https://studentlife.utoronto.ca/department/accessibility-services/`

English Language Instruction. For information on campus writing centres and writing courses, please visit

- `http://www.writing.utoronto.ca`.

Other Resources.

- Student Life Programs and Services `http://www.studentlife.utoronto.ca`
- Academic Success Centre `http://www.studentlife.utoronto.ca/asc`
- Health and Wellness Centre `http://www.studentlife.utoronto.ca/hwc`

AI Policy

Artificial Intelligence (AI) tools are becoming common in research, scientific communication, and technical work. In this course, the responsible use of AI is permitted and encouraged when it supports learning. However, AI should be used as a tool to assist your thinking, not as a substitute for your own understanding, creativity, or effort.
Students are expected to critically evaluate all AI-generated content: students are responsible for verifying the correctness of any mathematics, code, explanations, references, figures, or other material submitted.

Responsible use of AI includes acknowledging when it was used. If you use AI to assist with any submitted work in this course, please acknowledge it.

Final Project. For the final project, students must disclose how AI was used. The report and presentation should reflect the students' own understanding and contributions, not AI-generated content.

In this syllabus, AI was used to help with checking for grammar errors and typos, and refining the preamble.

Academic Integrity

Academic integrity is fundamental to learning and scholarship at the University of Toronto. Participating honestly, respectfully, responsibly, and fairly in this academic community ensures that the University of Toronto degree that you earn will be valued as a true indication of your individual academic achievement, and will continue to receive the respect and recognition it deserves.
Familiarize yourself with the University of Toronto's Code of Behaviour on Academic Matters. It is the rule book for academic behaviour at the University of Toronto, and you are expected to know the rules.

https://www.academicintegrity.utoronto.ca/

The University of Toronto treats cases of academic misconduct very seriously. All suspected cases of academic dishonesty will be investigated following the procedures outlined in the Code. The consequences for academic misconduct can be severe, including a failure in the course and a notation on your transcript. If you have any questions about what is or is not permitted in this course, please do not hesitate to contact your instructor or the course coordinator. If you have questions about appropriate research and citation methods, seek out additional information from your instructor or from other available campus resources like the University of Toronto Writing Website. If you are experiencing personal challenges that are having an impact on your academic work, please speak to your instructor or seek the advice of your college registrar.

Copyright Notice

You must not take audio recordings or video recordings of lectures or tutorials unless you received the written consent of the person whose work you are recording (consent will normally be given for accessibility reasons).
The Copyright Act protects all of the course materials such as slides, assignments, the lectures themselves and exams. They are yours to use and learn from; but you do not own them.